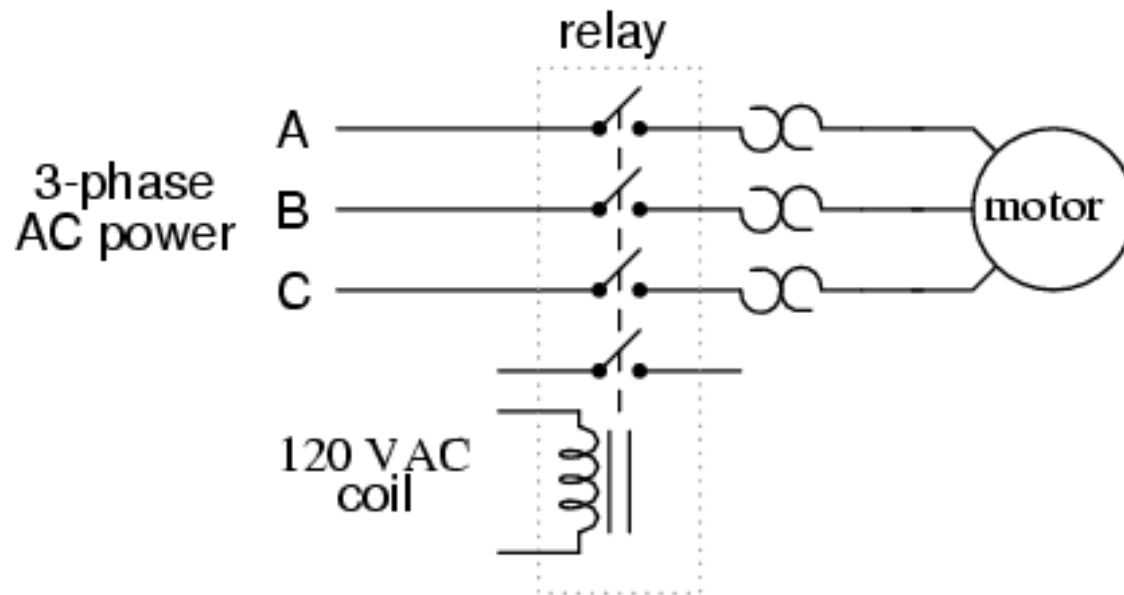


Contactors & Relays

Contactors

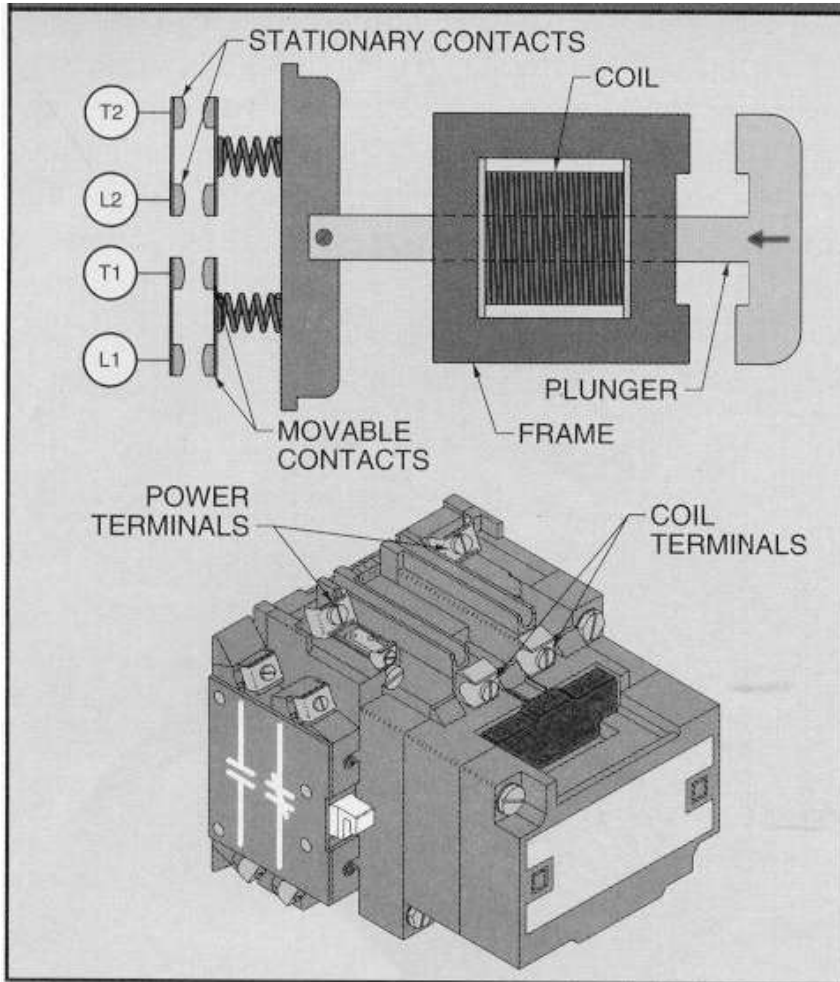
Contactors are relays that switch high current loads a.k.a magnetic starters



Manual Motor Starter



Contactors

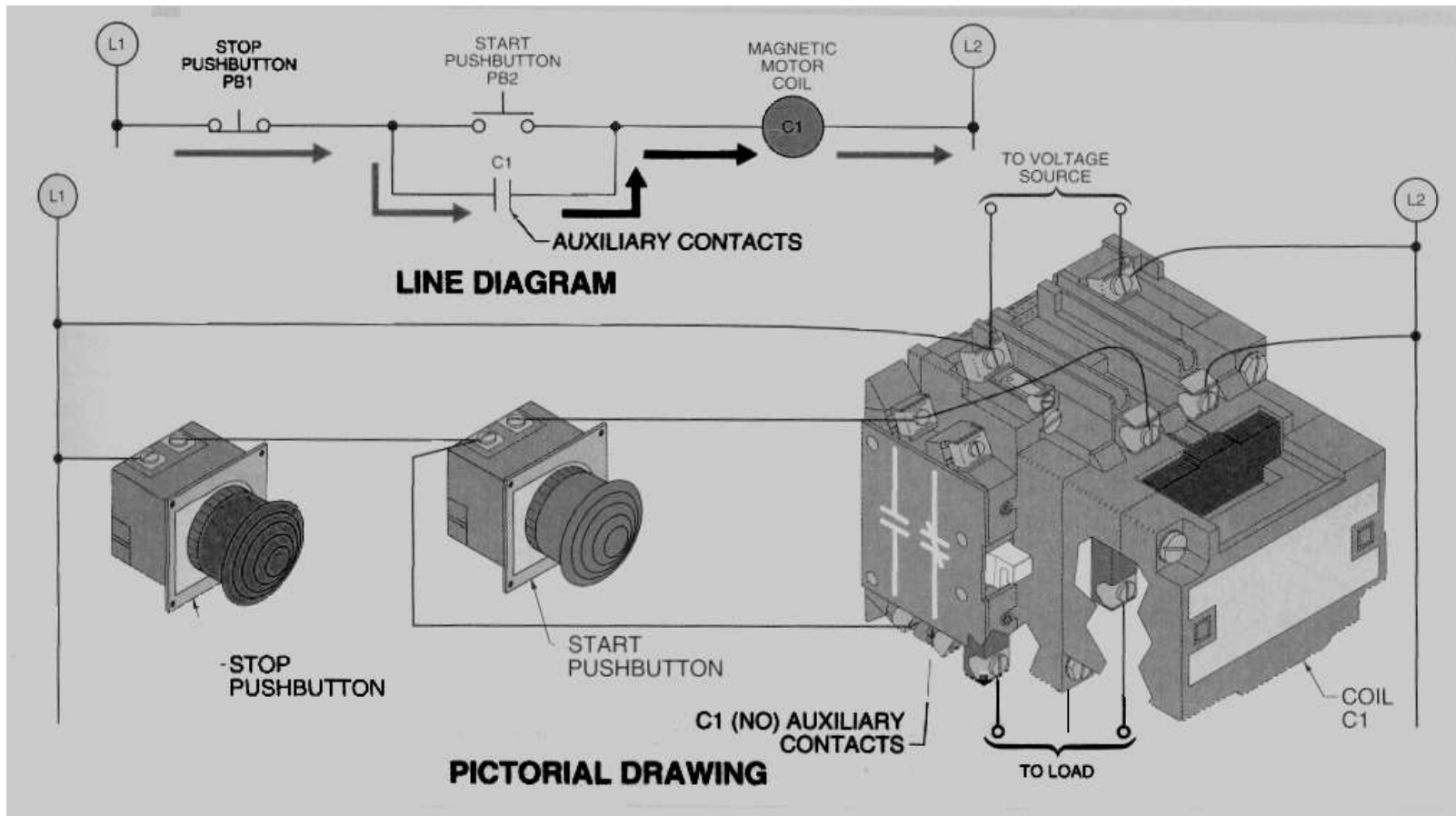


A **Contactor** is a control device that uses a small control current to energize or de-energize the load connected to it.

Abouts:

- A contactor has a frame, plunger, and a solenoid coil.
- The action of the plunger is used to close (or open) sets of contacts.
- A contactor does not include overload protection.
- The closing of the contacts allows electrical devices to be controlled from remote locations.

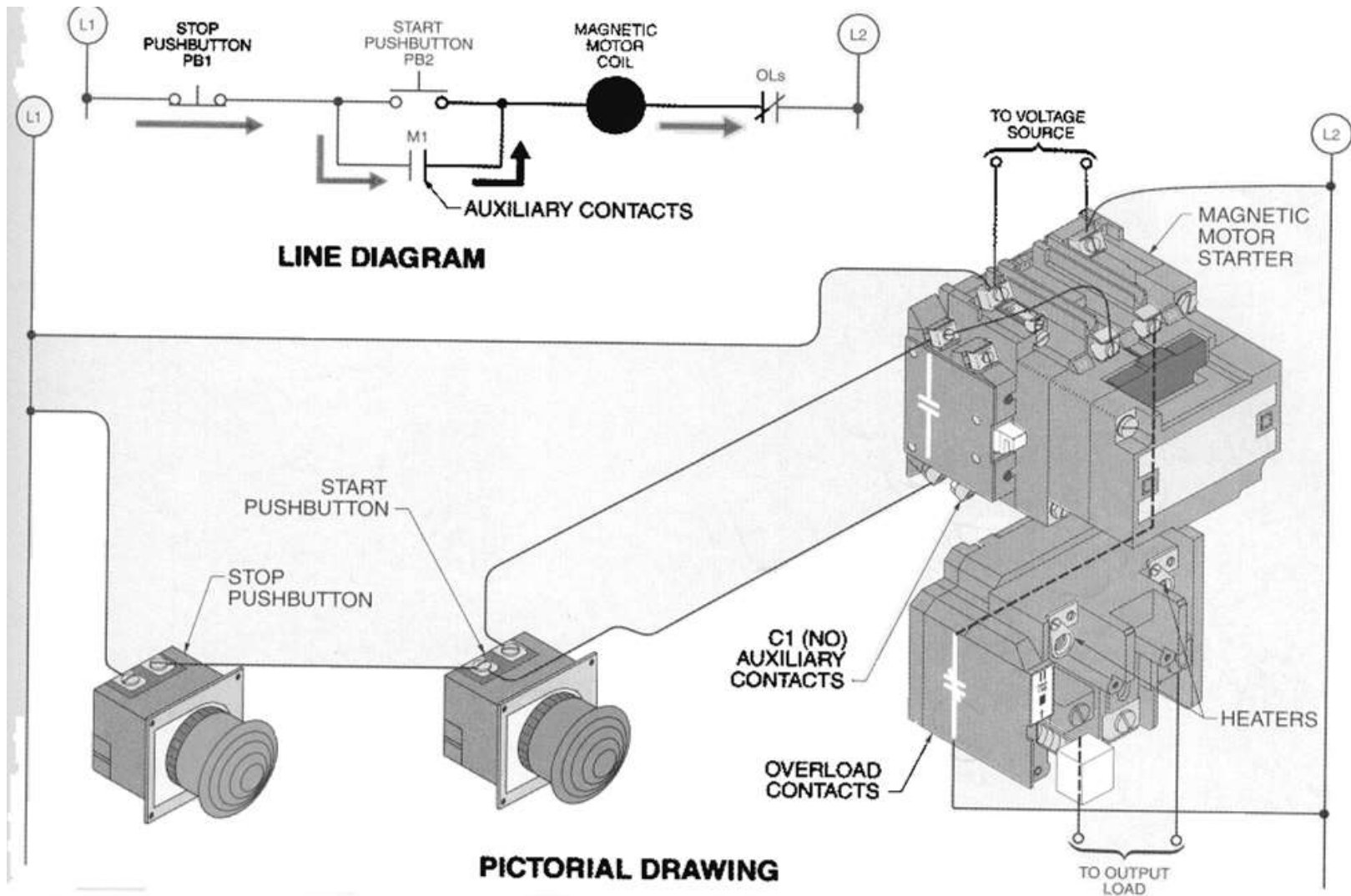
Example of a Wired Contactor



Magnetic Motor Starters

- A magnetic motor starter is an electrically-operated switch (contactor) that includes motor overload protection.
- Magnetic motor starters are identical to contactors except that they have **overloads** attached to them.
- The overloads have heaters or electronic overloads (located in the power circuit) which sense excessive current flow to the motor.
- The heaters open the NC overload contacts (located in the control circuit) when the overload becomes dangerous to the motor.

Magnetic Starter



Magnetic Motor Starter



L1 (1) - first line in from power source (phase 1 for 3ph / Neutral for 1ph)

L2 (3) - second line in from power source (phase 2 for 3ph / Hot for 1ph* see below for alternate wiring using L2 & L3)

L3 (5) - third line in from power source (phase 3 for 3ph / NC for 1ph)

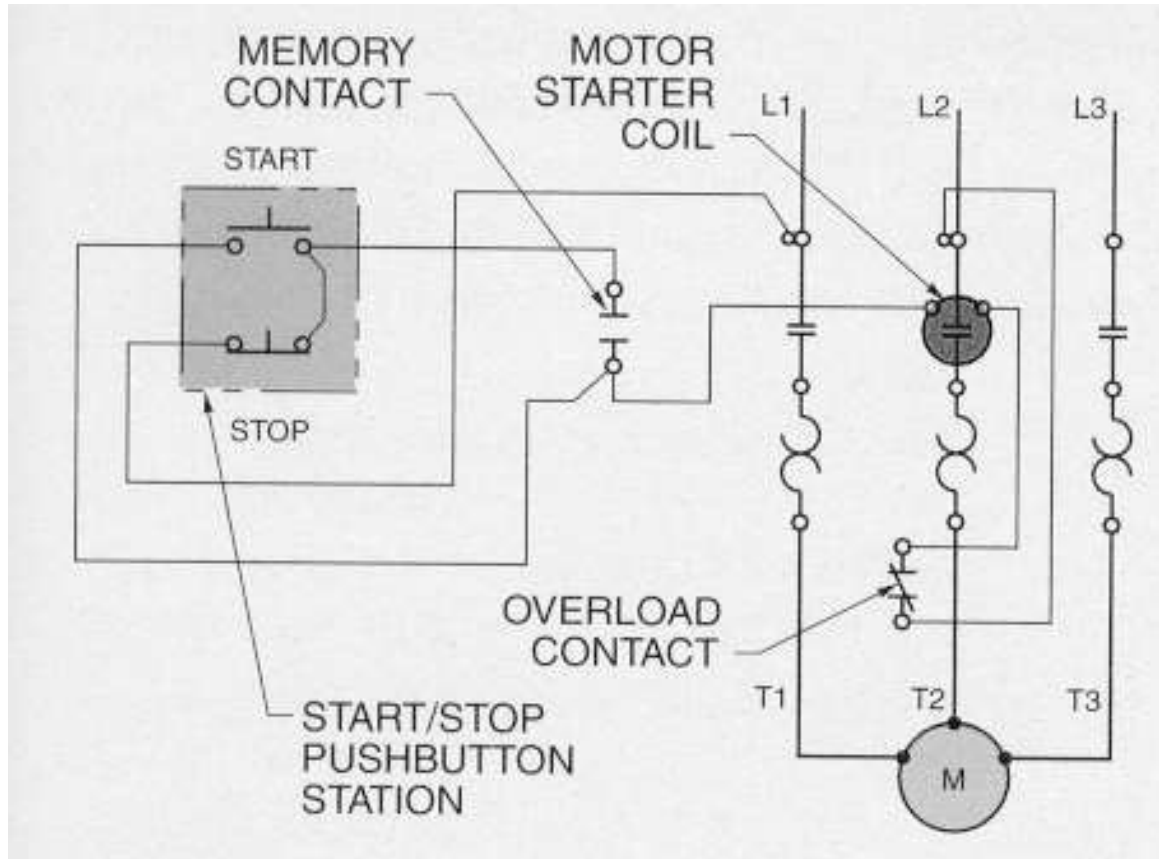
COIL

T1 (2) - first line out to motor (phase 1 for 3ph / Hot for 1ph)

T2 (4) - second line out to motor (phase 2 for 3ph / NC for 1ph* see below for alternate wiring)

T3 (6) - third line out to motor (phase 3 for 3ph / Neutral for 1ph)

Motor Starter Control Circuit



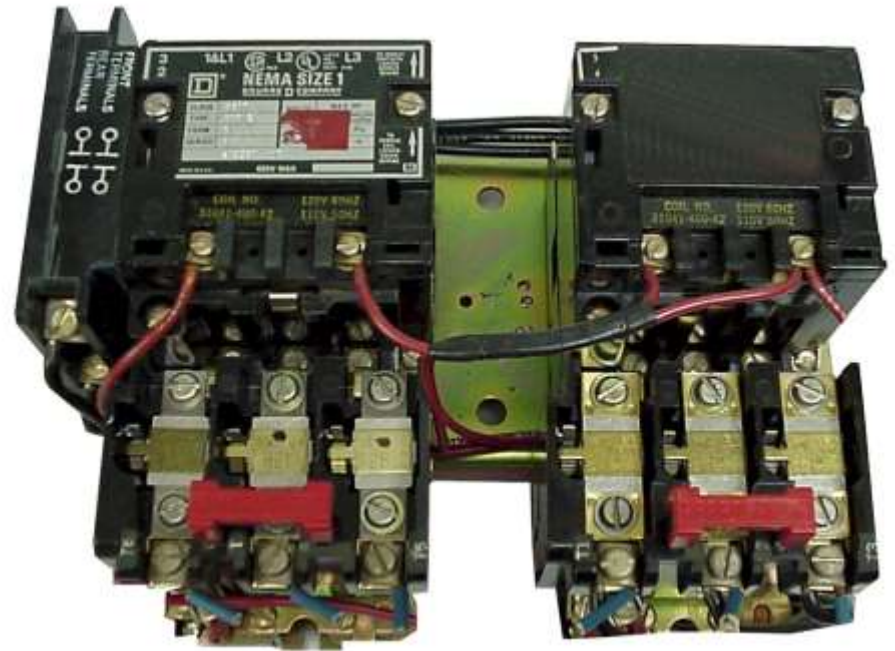
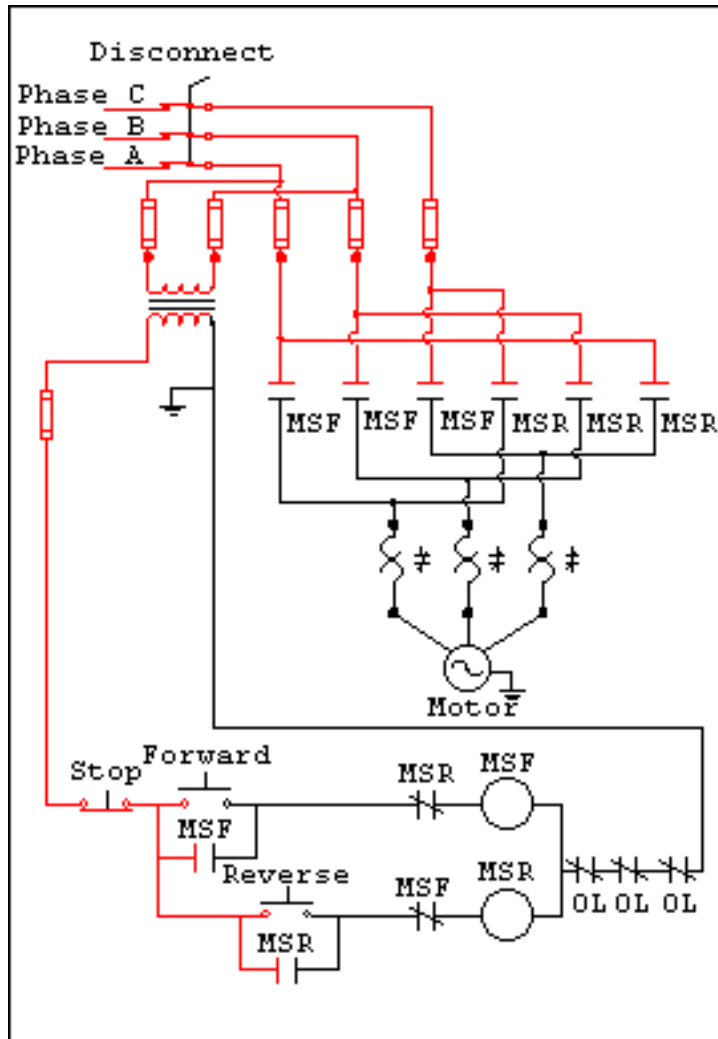
- Alternate method of drawing the electrical circuit

NEMA vs IEC

If we compare the NEMA magnetic motor starter to the IEC magnetic motor starter, the following differences would be noticed:

- An IEC device is physically smaller than a comparable NEMA device.
- An IEC device is usually less expensive than a comparable NEMA device.
- An IEC device has a life cycle of approximately one million operations while a comparable NEMA device has a life cycle of almost four times that number.
- An IEC device should normally be protected with fast-acting, current-limiting fuses while a NEMA device can be protected with conventional time delay fuses.

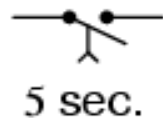
Reversing Starter



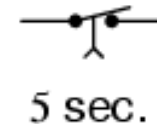
Relay Timers

- ON Delay

Normally-open, timed-closed

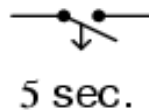


Normally-closed, timed-open

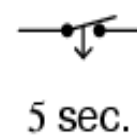


- OFF Delay

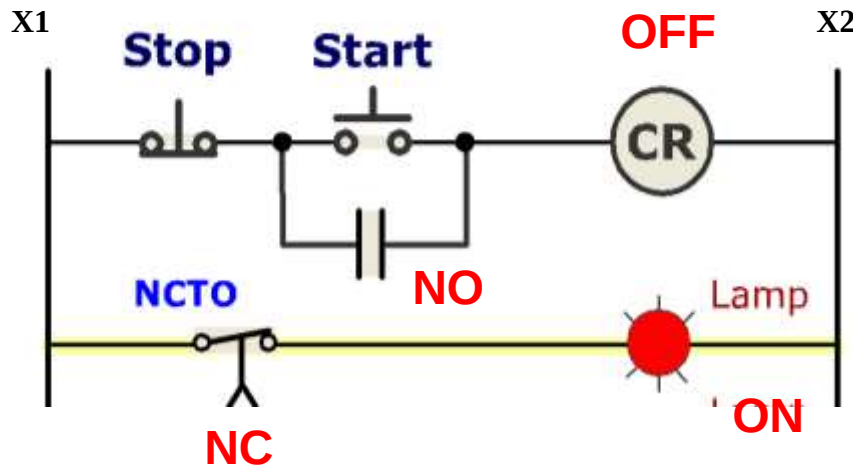
Normally-open, timed-open



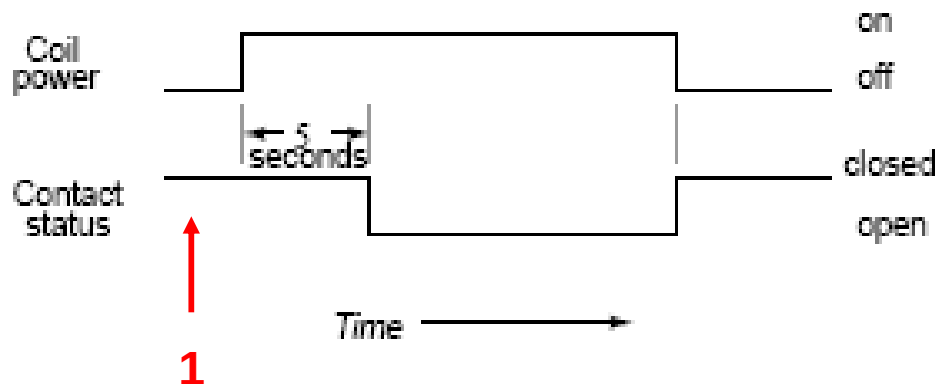
Normally-closed, timed-closed



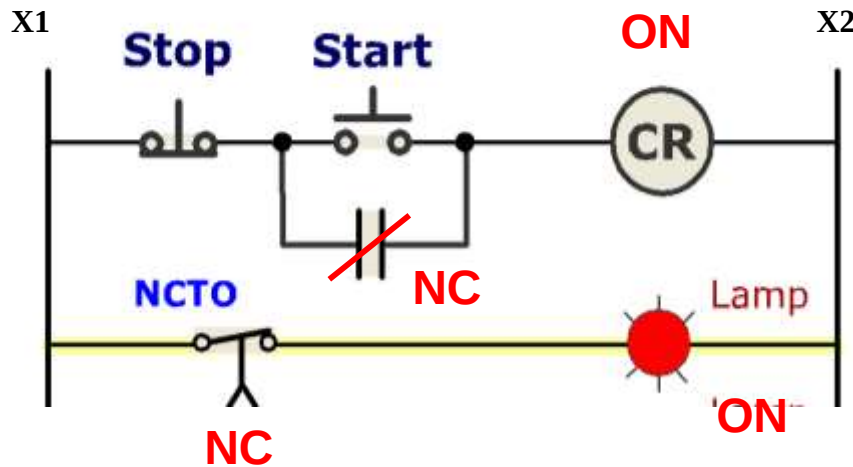
Wired ON Delay



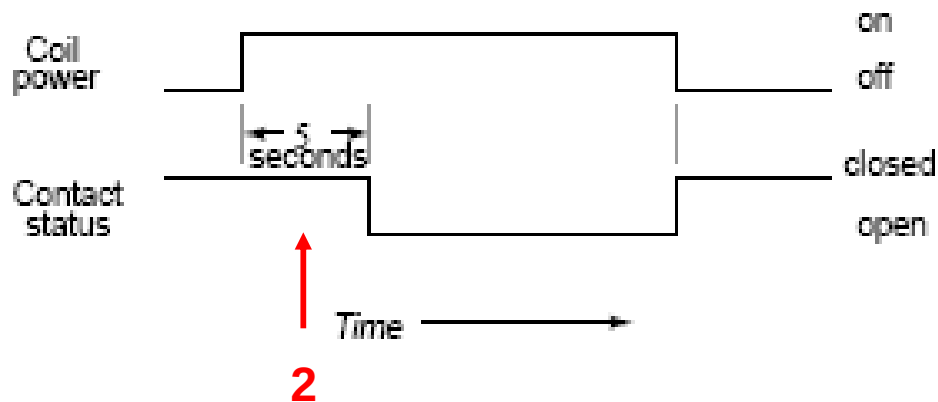
1. Energy applied to power rails



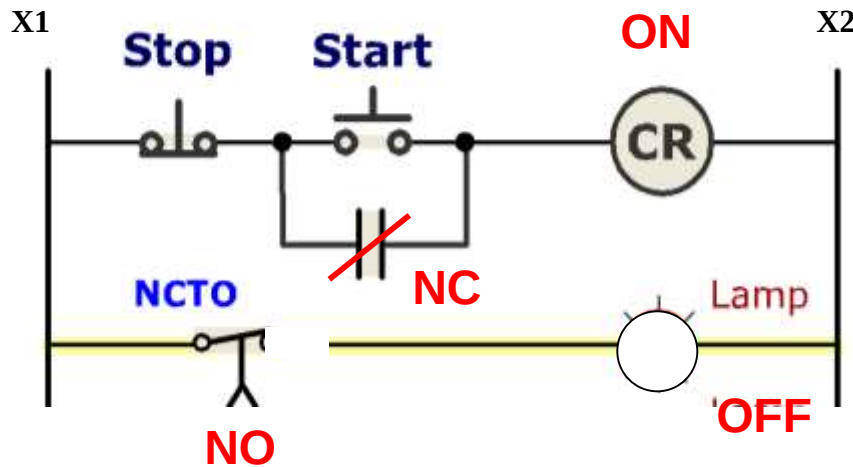
Wired ON Delay - NCTO



1. Energy applied to power rails
2. **Start PB is pressed**
 - Coil is energized
 - Holding contact close
 - Timer contact stays closed, lamp stays on.
 - Count begins (5 sec)



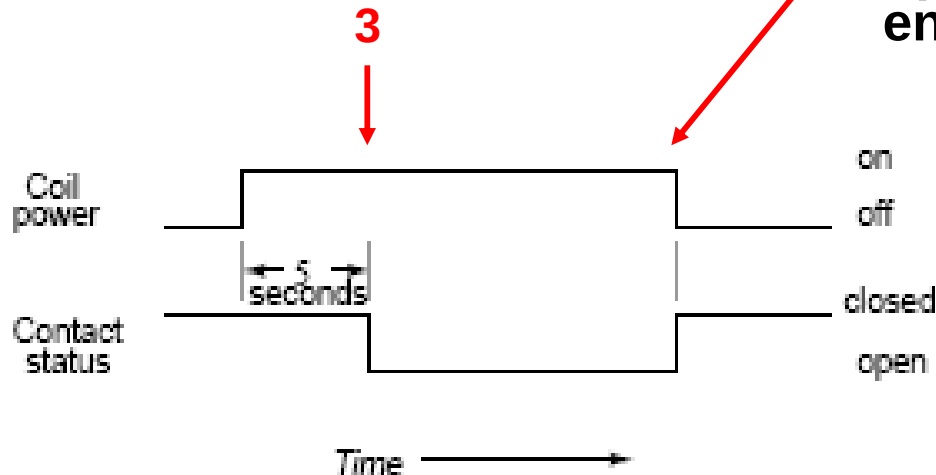
Wired ON Delay - NCTO



3. Timer count ends

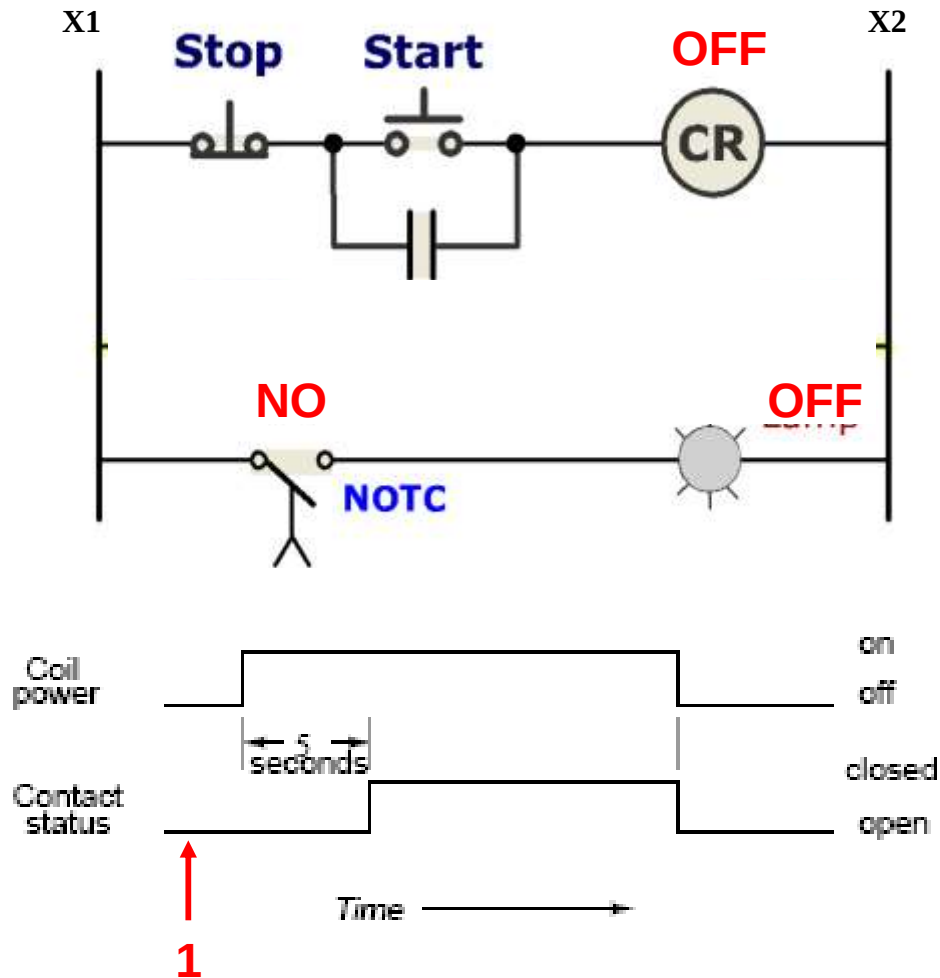
- Coil is still energized
- Timer contact open
- lamp goes off.

4. Timer contacts remain open until the coil is de-energized



The **N**ormally **C**losed contact will take 5 seconds **T**o **O**pen when the coil is energized.

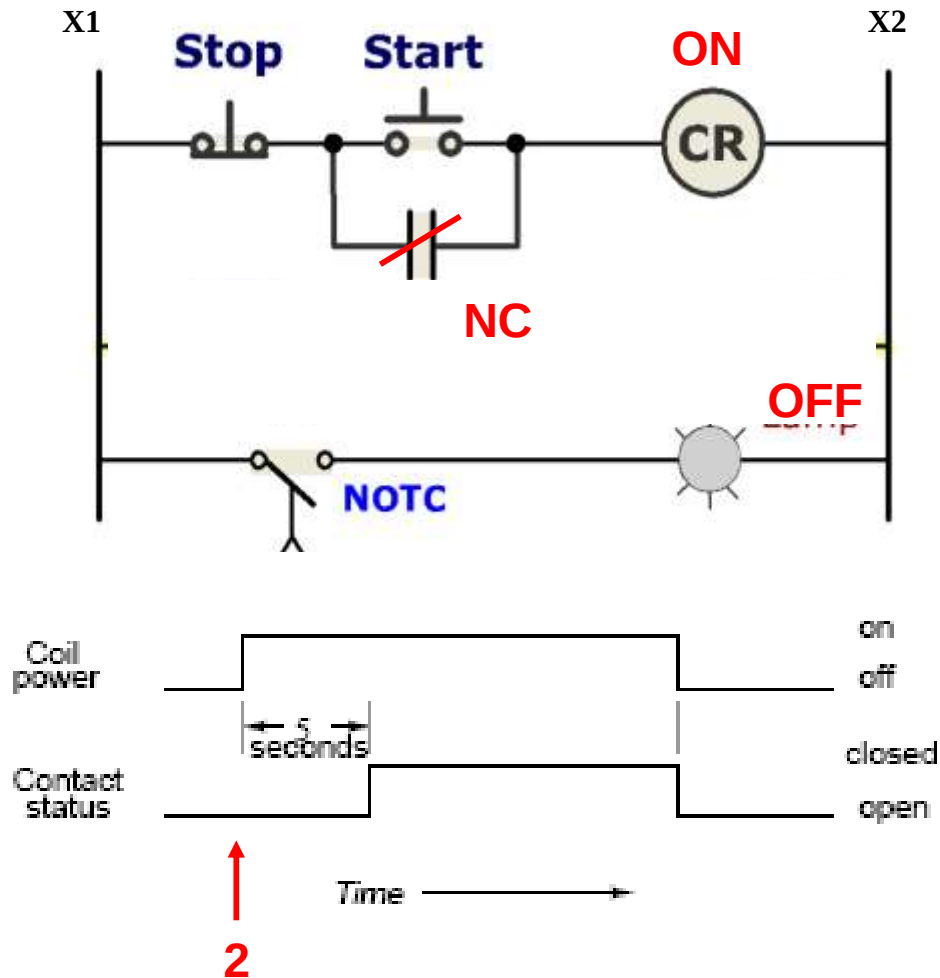
ON Delay - NOTC



1. Power is applied to rails

The **N**ormally **O**pen contact will take 5 seconds **T**o **C**lose when the coil is energized.

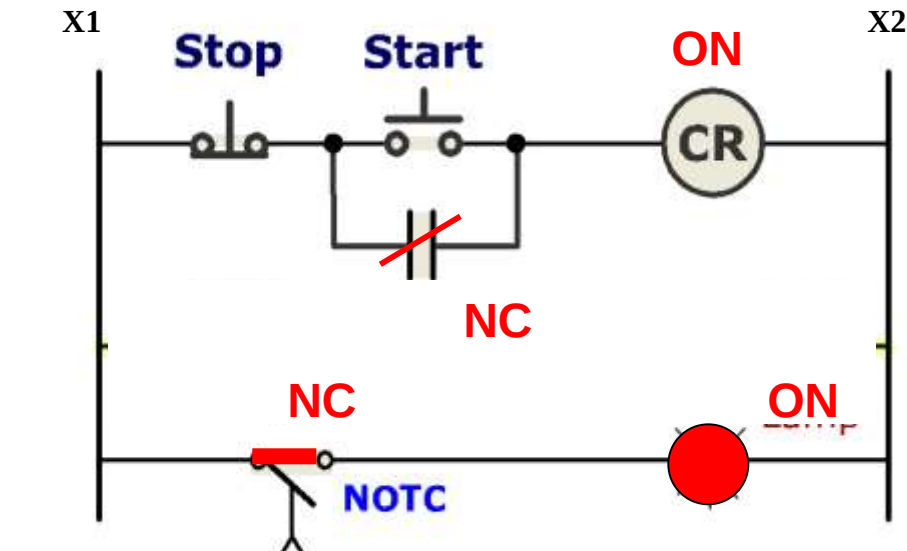
ON Delay - NOTC



2. Start PB is pressed
- Coil energizes
 - Holding contacts close
 - Timer contacts stay open
 - Lamp stays off
 - Counter starts to count (5 sec)

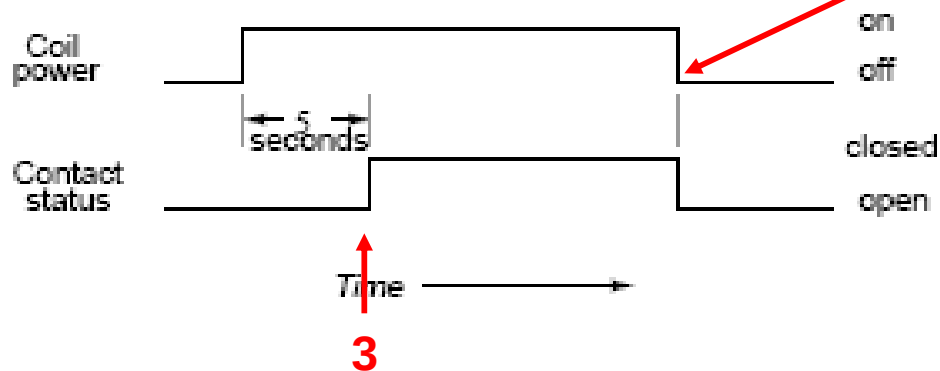
The **N**ormally **O**pen contact will take 5 seconds **T**o **C**lose when the coil is energized.

ON Delay - NOTC



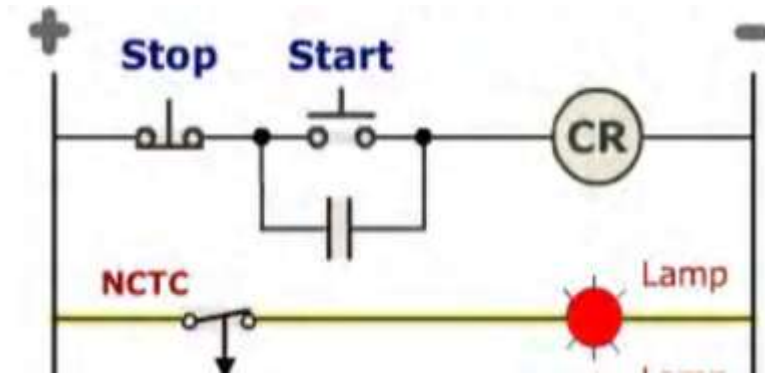
3. Counter finishes count
- Coil stays energized
 - Timer contacts close
 - Lamp goes on

4. Timer contacts will open when relay coil is de-energized.

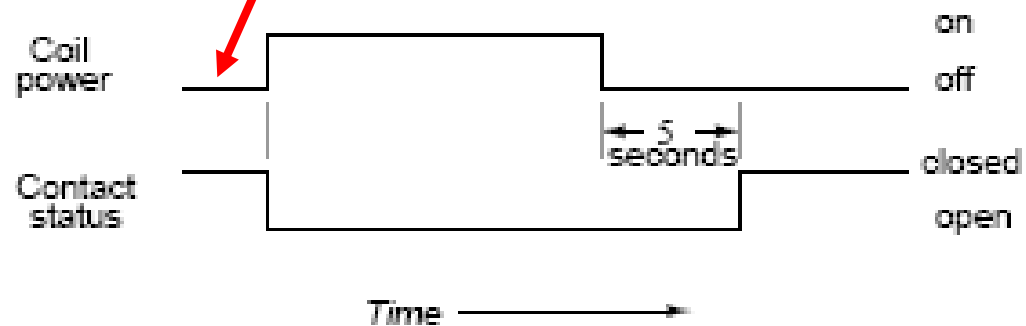


The **N**ormally **O**pen contact will take 5 seconds **T**o **C**lose when the coil is energized.

OFF Delay - NCTC

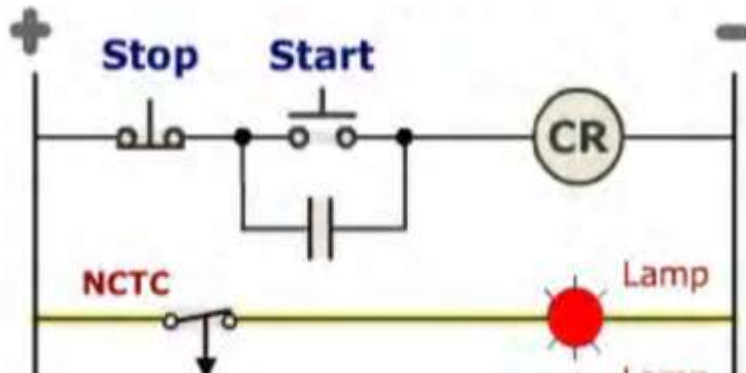


1. Power is applied to rails
2. Coil is off, contacts are closed, lamp is on

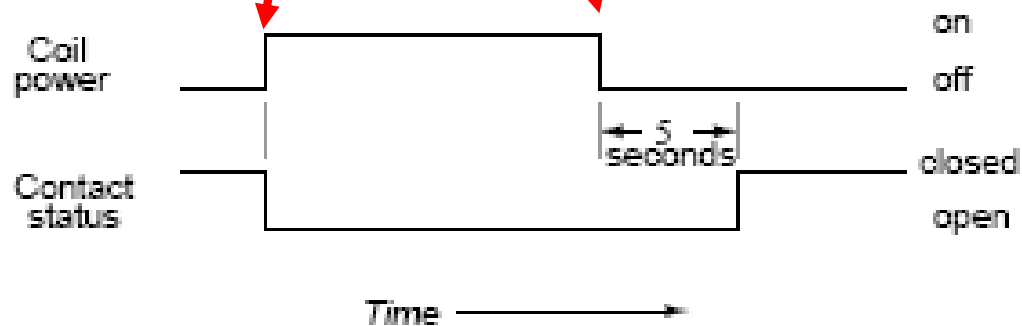


The timer contacts will close 5 seconds after the coil is de-energized

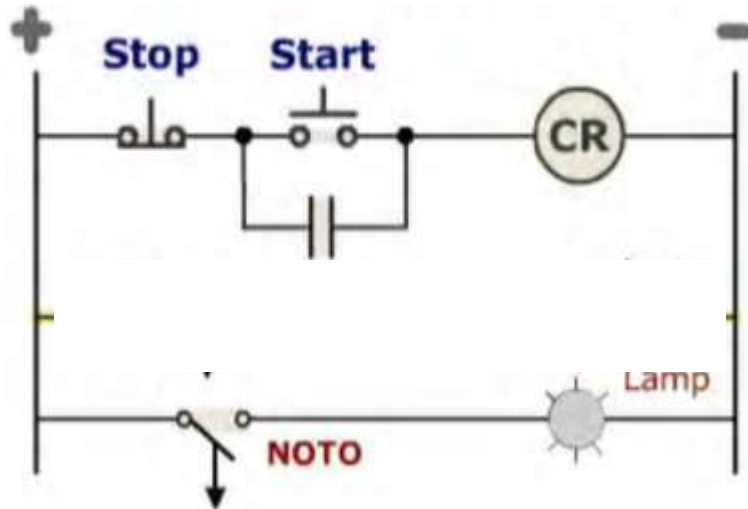
OFF Delay - NCTC



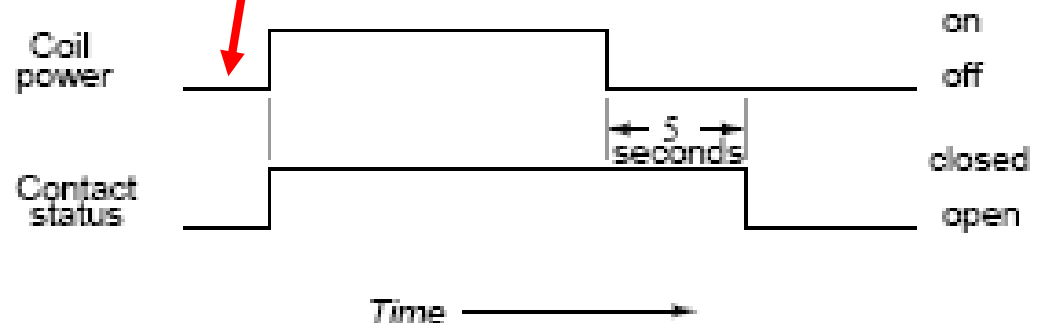
3. Start PB is pressed
4. Timer contacts open
5. Counter will start to count only when coil is de-energized.



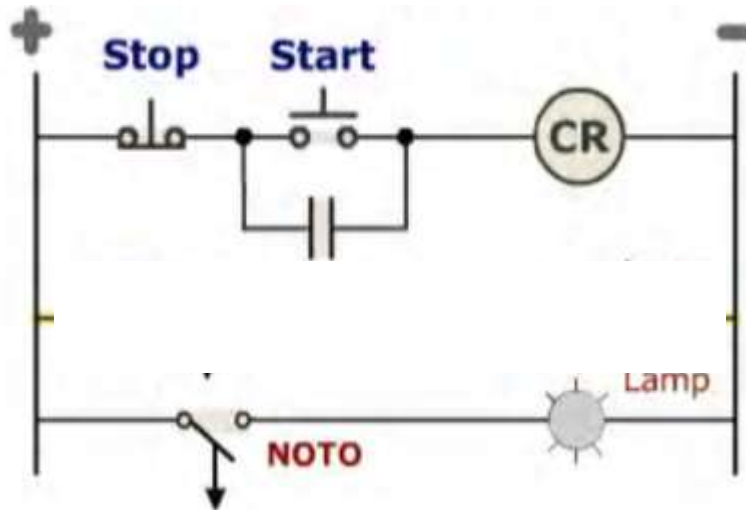
OFF Delay - NOTO



1. Power is applied to rails
2. Coil is off, contacts are closed, lamp is on



OFF Delay - NOTO



3. Start is pressed.
4. Contacts close, lamp on
5. Counter only starts when coil is de-energize

